

# Dually Flat Affine Geometry of the Probability Simplex Bases on Kaniadakis' Logarithm for Applications to Compositional Data Statistics

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**Abstract:** We use Kaniadakis' logarithm to mimic the standard construction of the affine statistical manifold. The parallel transport duality is conserved with an adaptation of the set-up consisting in the introduction of the escort probability functions and a new bilinear form on fibres. The use of exponential charts requires computations to be implemented.

**Keywords:** Information geometry; Compositional data; Statistical bundle; Kaniadakis entropy.

## 1. Introduction

This research is focused on a special kind of Kaniadakis statistics for Compositional data (CoDa). Compositional data is the data where all of a (row) vector's (i.e.  $[x_1, x_2, \dots, x_D]$ ) components are strictly positive real values and contain solely relative information; the composition is called a D-part composition [1]. Compositional data is often expressed as closed form and total to a fixed value, such as 1 for parts per unit or 100 for percentage measurements [2].

Compositional data is often found in geosciences and other scientific disciplines for classification, discrimination, and categorization. The use of CoDa analysis is closely related to geosciences and biology, where the data is mostly expressed as proportions or concentrations without mentioning the total size explicitly [3]. Significant advancement has been accomplished during the last thirty to forty years, and recently the term CoDa analysis has been employed to "Insist on the idea that the study goals or hypotheses, which place more of an emphasis on relative than absolute values, are what ultimately determine composition rather than the data, which may not be pieces of a whole or may not have a fixed sum" [4]. These qualities make CoDa analysis the most powerful tool for applications outside the tradition of hard sciences [5]. Current studies in management, economics, and social sciences have shown in practice the benefits of compositional methods in handling a wide variety of problems, which range from market shares, and customer segmentation to tourism, transport systems, financial ratios, and many more (see [4,6–8]).

Atchison [9] introduced CoDa analysis. The research conducted by the [5] on financial data used the Atchison methods to examine how European nations contribute to the total amount of systemic risk (SRISK). They find that the distinctive quality of CoDa analysis, especially the Atchison geometry, is very effective in determining the threats of possible instability concerns offered by smaller institutions and nations that might not completely emerge from the scale of their systemic risk. CoDa analysis, in addition, when compared to previous literature [10,11] they used size of systemic risk (SRISK) using either financial units: popular indicators expressed in financial units include, among others, SRISK and Covariance (CoVaR) based measures for comparative analysis.

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## 2. Compositional data

The Center for Risk Management at the University of Lausanne provides systemic risk assessments for European financial institutions, which we used in our empirical study using the above Kaniadakis methods (<http://www.crml.ch>). The dataset enables the determination of SRISK country-level values, a market-based systemic risk indicator first proposed in [12,13] and most recently examined in [14].

The characteristics of SRISK are popular in the literature, and particularly SRISK is used to recognize weak institutions and countries with a systemwide impact before a crisis occurs [10] and can help forecast real sector performance [15].

This literature has mainly focused on the non-negative values of SRISK. In this work, we focus on implementing the Kaniadakis methods to see the different European countries as a part of compositional data. We developed work started in [5], where they first introduced compositional data analysis to examine the distribution of relative contributions to SRISK connected with key European nations from 2008 to 2021.

### 2.1. Description of Data

In this research, we take the sample of ten European economies (Belgium, Denmark, France, Germany, Greece, Italy, Netherlands, Spain, Switzerland, UK) with annual SRISK measurements collected at the end of December for 2008–2021. Every number is stated in billions of euros. Like most CoDa method applications, the sample does not cover Europe. Therefore, the 10 components that make up our SRISK compositions are just a portion of all those that may be used. CoDa analysis, however, is predicated on the basic notion of sub-compositional coherence, which ensures that a compositional study conducted on a subset of components is consistent with the same analysis performed on the entire composition.

CoDa Terminology: CoDa point vs probability function. Refs about CoDa [9] [1].

Atchison geometry is an affine geometry on the simplex that could be described in a less parameter-dependent way as an Information Geometry. Refs about non-parametric IG. [16] and MS notes if available.

The key notion of exponential family admits generalizations, in particular Kaniadakis one. Refs. [17]? This presentation is described below.

We apply Kaniadakis formalism to a problem in Data Analysis in Finance. This application is compared with published research that uses conventional methods. In particular, note:

- the use of special divergences whose rational depends on the affine structure imposed on compositional data
- The use of the affine barycenter

*Move here the description of the data.*

The example discussed is intended to be both of independent interest in the field of Finance and an illustration of Compositional Data analysis made with Kaniadakis affine formalism. A short critical assessment is presented in the conclusion.

### 2.2. Kaniadakis' logarithm

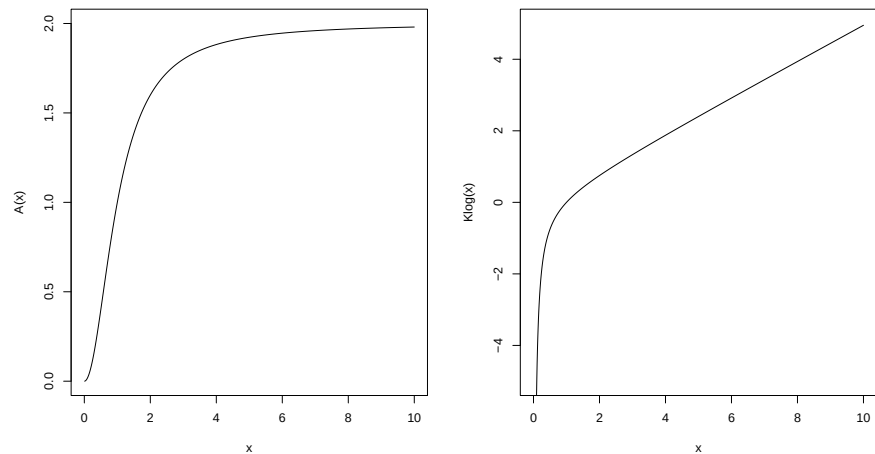
We focus on a special case of Kaniadakis' logarithm [18,19]. In the suggestive formalism introduced by Naudts [17],

$$\log_K(x) = \frac{1}{2} \left( x - \frac{1}{x} \right) = \int_1^x \frac{du}{A(u)}, \quad \text{with} \quad A(u) = \frac{2u^2}{1+u^2}, \quad (1)$$

$$\exp_K(y) = \log_K^{-1} y = y + \sqrt{1+y^2} = \exp \int_0^y \frac{dv}{\sqrt{1+v^2}}. \quad (2)$$

See ??.

The graph of  $y = \log_K(x)$  is a semi-algebraic function,



**Figure 1.** Naudt's function and Kaniadakis' logarithm.

$$x^2 - 2xy - 1 = 0, \quad x > 0.$$

The equation reduces any polynomial in  $y$  and  $x = \exp_K(y)$ , for example,

$$\begin{aligned} \exp_K(y)^2 &= 2y \exp_K(y) + 1, \\ \exp_K(y)^3 &= \exp_K(y)(2y \exp_K(y) + 1) = \dots, \end{aligned}$$

and so on. This is an algebraic feature, cf [20].

We list below the main known properties of the K-logarithm and the K-exponential. We have written some proofs in Appendix A for our and reader's convenience.

$$\log_K\left(\frac{1}{x}\right) = -\log_K(x), \quad \exp_K(y) \exp_K(-y) = 1. \quad (3)$$

$$\frac{d}{dy} \exp_K(y) = A(\exp_K(y)) = \frac{1}{\sqrt{1+y^2}} \exp_K(y) \quad (4)$$

$$\exp_K(x) \exp_K(y) = \exp_K\left(x\sqrt{1+y^2} + y\sqrt{1+x^2}\right) \quad (5)$$

$$\log_K(x_2) - \log_K(x_1) < \frac{1}{A(x_1)}(x_2 - x_1), \quad x_1 \neq x_2 \quad (6)$$

$$\exp_K(y_2) - \exp_K(y_1) > A(\exp_K(y_1))(y_2 - y_1), \quad y_1 \neq y_2 \quad (7)$$

### 2.3. K-exponential form of a positive probability density

If  $\Omega$  is a finite sample space, the probability simplex on  $\Omega$  is  $\mathcal{P}(\Omega)$  and the open probability simplex is  $\mathcal{E}(\Omega)$ .

For all  $p \in \mathcal{E}(\Omega)$ , the function  $A \circ p$ ,  $A$  as in eq. (1), is strictly positive and provides a positive weight on  $\Omega$ . It is proportional, but usually not equal, to a probability function,

$$\tilde{p} \propto A \circ p, \quad \text{namely,} \quad \tilde{p}(x) = \frac{A(p(x))}{\sum_{x \in \Omega} A(p(x))}.$$

We will also write  $\tilde{p} = A \circ p / \overline{A \circ p}$ . The mapping  $\mathcal{E}(\Omega) \ni p \mapsto \tilde{p} \in \mathcal{E}(\Omega)$  is the *escort* mapping, see [17]. See [21, §3.1] for a discussion of injectivity and surjectivity. We introduce a symbol for the escort expectation.  $\mathbb{E}_p[u] = \sum_{x \in \Omega} u(x) \tilde{p}(x)$ .

For  $p, q \in \mathcal{E}(\Omega)$ , we define the *Kaniadakis divergence* as

$$\begin{aligned}
\tilde{D}(p|q) &= \tilde{\mathbb{E}}_p[\log_K(p) - \log_K(q)] \\
&= \sum_{x \in \Omega} (\log_K(p) - \log_K(q)) A \circ p / \overline{A \circ p} . \\
&= \sum_{x \in \Omega} (\log_K(p) - \log_K(q)) \tilde{p}(x)
\end{aligned} \tag{8}$$

Clearly,  $\tilde{D}(p|p) = 0$ . If  $p \neq q$ , from the concavity in eq. (6),

$$\begin{aligned}
\tilde{D}(p|q) &= (\overline{A \circ p})^{-1} \sum_{x \in \Omega} A(p(x)) (\log_K(p(x)) - \log_K(q(x))) \\
&> (\overline{A \circ p})^{-1} \sum_{x \in \Omega} (p(x) - q(x)) \\
&= 0 .
\end{aligned}$$

Fix  $p \in \mathcal{E}(\Omega)$ . For all  $q \in \mathcal{E}(\Omega)$ , define

$$\begin{aligned}
s_p(q) &= (\log_K(q) - \log_K(p)) + \tilde{D}(p|q) = (\log_K(q) - \log_K(p)) - \tilde{\mathbb{E}}_p[(\log_K(q) - \log_K(p))] , \\
\text{then, for } u &= s_p(q),
\end{aligned}$$

$$q = \exp_K(u - \tilde{D}(p|q) + \log_K(p)) , \quad \tilde{\mathbb{E}}_p[u] = 0 . \tag{9}$$

Conversely, for all  $p \in \mathcal{E}(\Omega)$ , if  $u$  is a random variable such that  $\tilde{\mathbb{E}}_p[u] = 0$ , the real function

$$\mathbb{R}_+ \ni \psi \mapsto \sum_{x \in \Omega} \exp_K(u(x) - \psi + \log_K(p(x)))$$

is continuous, goes to 0 as  $\psi \rightarrow \infty$ , and, for  $\psi = 0$ , takes a value larger than 1 because of eq. (7):

$$\begin{aligned}
\sum_{x \in \Omega} \exp_K(u(x) + \log_K(p(x))) &> \sum_{x \in \Omega} \exp_K(\log_K(p(x))) + \sum_{x \in \Omega} A(\exp_K(\log_K(p(x))))u(x) \\
&= \sum_{x \in \Omega} p(x) + \sum_{x \in \Omega} A(p(x))u(x) \\
&= 1 .
\end{aligned}$$

In conclusion, there exists a function,

$$K_p: S_p = \left\{ u \in L(\Omega) \mid \tilde{\mathbb{E}}_p[u] = 0 \right\} \ni u \mapsto K_p(u) \geq 0 ,$$

and  $K_p(u) > 0$  provided  $u \neq 0$ , such that

$$q = \exp_K(u - K_p(u) + \log_K(p)) \in \mathcal{E}(\Omega) ,$$

hence  $K_p(u) = \tilde{D}(p|q)$ . The mapping

$$s_p: \mathcal{E}(\Omega) \rightarrow S_p$$

is a bijection with inverse

$$e_p: S_p \ni u \mapsto \exp_K(u - K_p(u) + \log_K(p)) \in \mathcal{E}(\Omega) . \tag{10}$$

## 2.4. Properties of the cumulant function $K_p$ .

Let us compute the derivatives of the function  $K_p$ . We use a square bracket notation for the direction,

$$dK_p(u)[h] = \lim_{\theta \rightarrow 0} \theta^{-1} (K_p(u + \theta h) - K_p(u)) = \left. \frac{d}{d\theta} K_p(u + \theta h) \right|_{\theta=0}.$$

From eq. (10),

$$\begin{aligned} 0 &= \left. \frac{d}{d\theta} \sum_{x \in \Omega} \exp_K(u(x) + \theta h(x) - K_p(u + \theta h) + \log_K(p(x))) \right|_{\theta=0} \\ &= \sum_{x \in \Omega} A(\exp_K(u(x) + \theta h(x) - K_p(u + \theta h) + \log_K(p(x)))) (h(x) - \left. \frac{d}{dt} K_p(u + \theta h) \right|_{\theta=0}) \\ &= \sum_{x \in \Omega} A(\exp_K(u(x) - K_p(u) + \log_K(p(x)))) (h(x) - dK_p(u)[h]) \\ &= \sum_{x \in \Omega} A(e_p(u)) (h - dK_p(u)[h]). \end{aligned}$$

It follows that for each  $p \in \mathcal{E}(\Omega)$ , and  $u, h \in S_p$ , it holds

$$dK_p(u)[h] = \tilde{\mathbb{E}}_q[h], \quad \text{where } q = e_p(u) \text{ see eq. (10)}. \quad (11)$$

**Remark 1.** Similarly, one should compute  $d^2 K_p$  and the convex conjugate of  $K_p$ . See below for the duality and see [22], [23], and M. S. notes.

## 2.5. Bibliographical notes

Kaniadakis logarithm and exponential were first introduced in Kaniadakis [24]; Kaniadakis [25]. The application to IG used here appeared in Pistone [26]; Montrucchio and Pistone [27]; Montrucchio and Pistone [21]. These papers discuss both the finite state space and the general state space.

## 3. Affine space

**Remark 2.** The non-parametric affine geometry of the open probability simplex is a variation of the standard case; see [23] and MS notes. The main difference is the substitution of the expectation with the escort expectation. Some details remain for the moment unsettled.

### 3.1. Statistical bundle

The statistical bundle is an expression of the tangent space of  $\mathcal{E}(\Omega)$  as a dually flat affine statistical manifold in the sense of [28].

The statistical bundle  $S\mathcal{E}(\Omega)$  and each fiber  $S_q\mathcal{E}(\Omega)$  are defined by

$$S\mathcal{E}(\Omega) = \left\{ (q, v) \mid q \in \mathcal{E}(\Omega), \tilde{\mathbb{E}}_q[v] = 0 \right\}, \quad (12)$$

$$S_q\mathcal{E}(\Omega) = \left\{ v \in L(\Omega) \mid \tilde{\mathbb{E}}_q[v] = 0 \right\}, \quad q \in \mathcal{E}(\Omega). \quad (13)$$

In our setup, each fibre is a finite-dimensional vector space and can be identified with its dual. However, it is convenient to distinguish the two statistical bundles. The previous one is called *exponential* statistical bundle, while the *mixture* statistical bundle is

$$^*S\mathcal{E}(\Omega) = \left\{ (q, v) \mid q \in \mathcal{E}(\Omega), \tilde{\mathbb{E}}_q[v] = 0 \right\}, \quad (14)$$

$$^*S_q\mathcal{E}(\Omega) = \left\{ v \in L(\Omega) \mid \tilde{\mathbb{E}}_q[v] = 0 \right\}, \quad q \in \mathcal{E}(\Omega). \quad (15)$$

For each couple  $p, q \in \mathcal{E}(\Omega)$ , the mapping

$${}^e\mathbb{U}_p^q: S_p \mathcal{E}(\Omega) \ni v \mapsto v - \tilde{\mathbb{E}}_q[v] \in S_q \mathcal{E}(\Omega) \quad (16)$$

is a bijection. The  ${}^e\mathbb{U}_p^p$  is the identity mapping, and

$${}^e\mathbb{U}_q^r {}^e\mathbb{U}_p^q = {}^e\mathbb{U}_p^r,$$

The co-cycle of mappings  $({}^e\mathbb{U}_p^q)_{p,q}$  is the *exponential parallel transport* of the exponential statistical bundle.

The mapping defined for all  $p \in \mathcal{E}(\Omega)$ ,  $v \in {}^*S_p \mathcal{E}(\Omega)$ , and  $w \in S_p \mathcal{E}(\Omega)$  by

$$g: (p, v, w) \mapsto g_p(v, w) = \langle v, w \rangle_{A \circ p} = \sum_{x \in \Omega} A(p(x))u(x)v(x) = \overline{A \circ p} \tilde{\mathbb{E}}_p[vw]$$

provides a duality between the fibres of  $S \mathcal{E}(\Omega)$  and  ${}^*S \mathcal{E}(\Omega)$ .

Let us compute the dual of the exponential transport. For  $p, q \in \mathcal{E}(\Omega)$ ,  $v \in {}^*S_q \mathcal{E}(\Omega)$ , and  $w \in S_p \mathcal{E}(\Omega)$ ,

$$\begin{aligned} \langle v, {}^e\mathbb{U}_p^q w \rangle_{A \circ q} &= \sum_{x \in \Omega} A \circ q \, v(w - \tilde{\mathbb{E}}_q[w]) \\ &= \sum_{x \in \Omega} A \circ q \, vw - \tilde{\mathbb{E}}_q[w] \sum_{x \in \Omega} A \circ q v \\ &= \sum_{x \in \Omega} A \circ q \, vw \\ &= \sum_{x \in \Omega} A \circ p \left( \frac{A \circ q}{A \circ p} u \right) w \\ &= \left\langle \frac{A \circ q}{A \circ p} u, w \right\rangle_{A \circ p}. \end{aligned}$$

Now,  $\frac{A \circ q}{A \circ p} v \in {}^*S_p \mathcal{E}(\Omega)$ , hence the dual of the exponential transport is the *mixture transport*,

$${}^*({}^e\mathbb{U}_q^p)v = {}^m\mathbb{U}_q^p v = \frac{A \circ q}{A \circ p} v. \quad (17)$$

### 3.2. Velocity and auto-parallel curves

The following computation is a version of the original argument about Fisher's score. Let  $t \mapsto q(t) \in \mathcal{E}(\Omega)$  be a one-dimensional parametric model, a curve in geometric language. We assume it is smooth as a mapping in the vector space  $L(\Omega)$ . For each random variable  $f \in L(\Omega)$ ,

$$\begin{aligned} \frac{d}{dt} \mathbb{E}_{q(t)}[f] &= \sum_{x \in \Omega} f(x) \dot{q}(x; t) \\ &= \sum_{x \in \Omega} f(x) \frac{\dot{q}(x; t)}{A(q(x; t))} A(q(x; t)) \\ &= \sum_{x \in \Omega} f(x) \frac{d}{dt} \log_K(q(x; t)) A(q(x; t)) \\ &= \sum_{x \in \Omega} A \circ q(t) (f - \mathbb{E}_{\tilde{q}(t)}[f]) \frac{d}{dt} \log_K(q(t)) \\ &= \left\langle f - \mathbb{E}_{\tilde{q}(t)}[f], \frac{d}{dt} \log_K(q(t)) \right\rangle_{A \circ q(t)}. \end{aligned}$$

If we define the *velocity* of the curve to be

$$\dot{q}(t) = \frac{d}{dt} \log_K(q(t)) = \frac{\dot{q}(t)}{A(q(t))}. \quad (18)$$

we can check that  $\dot{q}(t) \in S_{q(t)} \mathcal{E}(\Omega)$  and  $(f - \mathbb{E}_{\tilde{q}(t)}[f]) \in {}^*S_{q(t)} \mathcal{E}(\Omega)$ . And symmetrically.  
The Cramer-Rao bound is

$$\begin{aligned} \left( \frac{d}{dt} \mathbb{E}_{q(t)}[f] \right)^2 &= \overline{A \circ q(t)}^2 \left\langle f - \mathbb{E}_{\tilde{q}(t)}[f], \frac{d}{dt} \log_K(q(t)) \right\rangle_{\tilde{q}(t)} \\ &\leq \overline{A \circ q(t)}^2 \mathbb{E}_{\tilde{q}(t)}[(f - \mathbb{E}_{\tilde{q}(t)}[f])^2] \mathbb{E}_{\tilde{q}(t)} \left[ \left( \frac{\dot{q}(t)}{A \circ q(t)} \right)^2 \right] \\ &= \sum_{x \in \Omega} A \circ q(t) (f - \tilde{\mathbb{E}}_q[f])^2 \sum \frac{\dot{q}(t)^2}{A \circ q(t)}. \end{aligned}$$

**Remark 3.** Notice that the variation computed with the escort probability function, namely  $f - \tilde{\mathbb{E}}_q[f]$  appears in ?? as a gradient of the expectation  $f \mapsto \mathbb{E}_q[f]$ .

A curve  $t \mapsto q(t)$  is *auto-parallel* for the mixture transport if

$$\mathfrak{m}\mathbb{U}_{q(t)}^{q(s)} \dot{q}(t) = \dot{q}(s).$$

For  $q(0) = q_0$  and  $q(1) = q_1$ ,

$$\frac{\dot{q}(0)}{A \circ q_0} = \frac{A \circ q(t)}{A \circ q_0} \frac{\dot{q}(t)}{A \circ q(t)} = \frac{1}{A \circ q_0} \dot{q}(t),$$

so that

$$q(t) = q_0 + t(q_1 - q_0).$$

The auto-parallel curves for the mixture transport are the ordinary geodesics of the simplex.

Let us compute the auto-parallel curves for the exponential transport,

$$\mathfrak{e}\mathbb{U}_{q(t)}^{q(s)} \dot{q}(t) = \dot{q}(s).$$

For  $q(0) = q_0$  and  $q(1) = q_1$ ,

$$\dot{q}(0) = \dot{q}(t) - \tilde{\mathbb{E}}_{q_0}[\dot{q}(t)] = \frac{d}{dt} \log_K(q(t)) - \frac{1}{A \circ q_0} \sum A \circ q_0 \frac{d}{dt} \log_K(q(t)),$$

so that, for some function  $\psi$

$$\log_K(q(t)) = \log_K(q_0) + t\dot{q}(0) - \psi(t).$$

Comparing with eq. (10), we have that the auto-parallel curve for the exponential transport is

$$q(t) = \exp_K(t\dot{q}(0) - K_{q_0}(t\dot{q}(0)) + \log_K(q_0)).$$

As seen above,

$$\dot{q}(0) = \log_K(q_1) - \log_K(q_0) - \tilde{\mathbb{E}}_{q_0}[\log_K(q_1) - \log_K(q_0)] = \log_K(q_0) + \tilde{D}(q_0|q_1).$$

### 3.3. Displacement

The machinery introduced above allows for the explicit definition of the affine structure in the sense of Weyl [29].

#### DEFINITIONS

Mixture displacement

$$\eta_p(q) = \frac{q - p}{A \circ p} \quad (19)$$

Exponential displacement

$$s_p(q) = (\log_K(q) - \log_K(p)) - \tilde{\mathbb{E}}_p[(\log_K(q) - \log_K(p))] \quad (20)$$

Both define affine coordinates (check it)

DEFINITION OF an AFFINE ATLAS on the statistical bundle

### 3.4. Velocity and acceleration

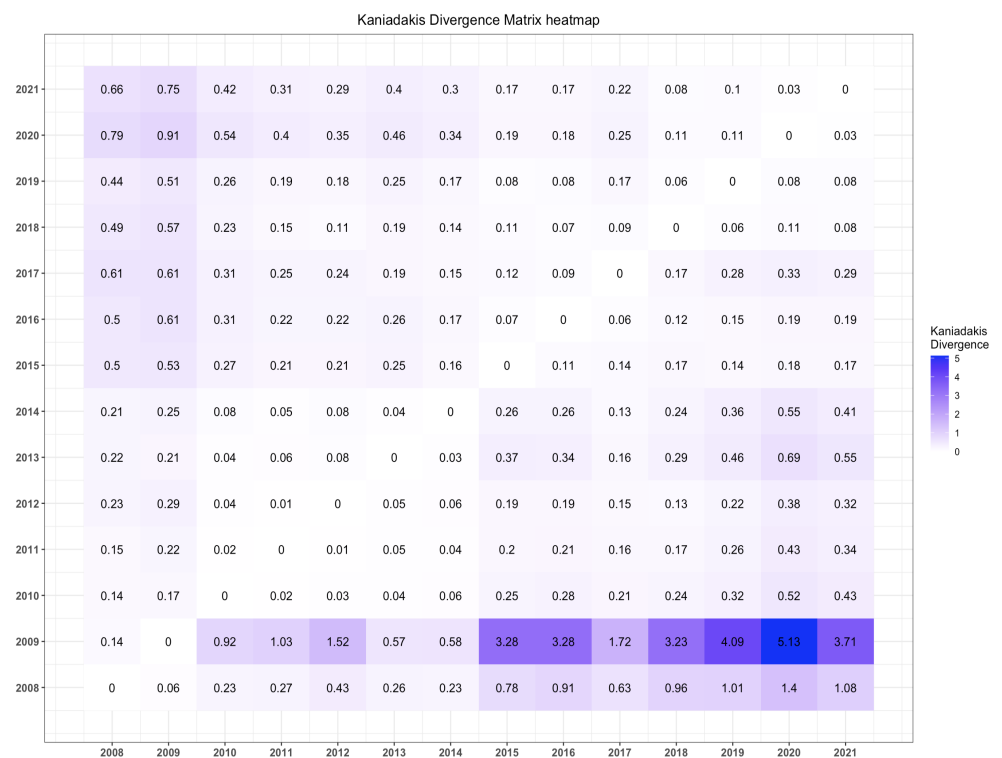
### 3.5. Gradient

### 3.6. Bibliographical notes

### 3.7. Data Analysis

#### 3.7.1. Kaniadakis divergence

First, we compute the Kaniadakis divergence defined in equation (8). Each point in  $(i, j)$  in Figure (2) is the Kaniadakis divergence of (CoDa point) corresponding to the year in the  $i^{th}$  row with respect to the (CoDa point) for the year in the  $j^{th}$  column. For example, the Kaniadakis divergence between 2020 and 2021  $K(2008|2009) = 0.14$ . Most values are smaller than one, except when the reference distribution is corresponding to the years 2008 or 2009 for the most recent years. The year 2009 deviates significantly from other years.



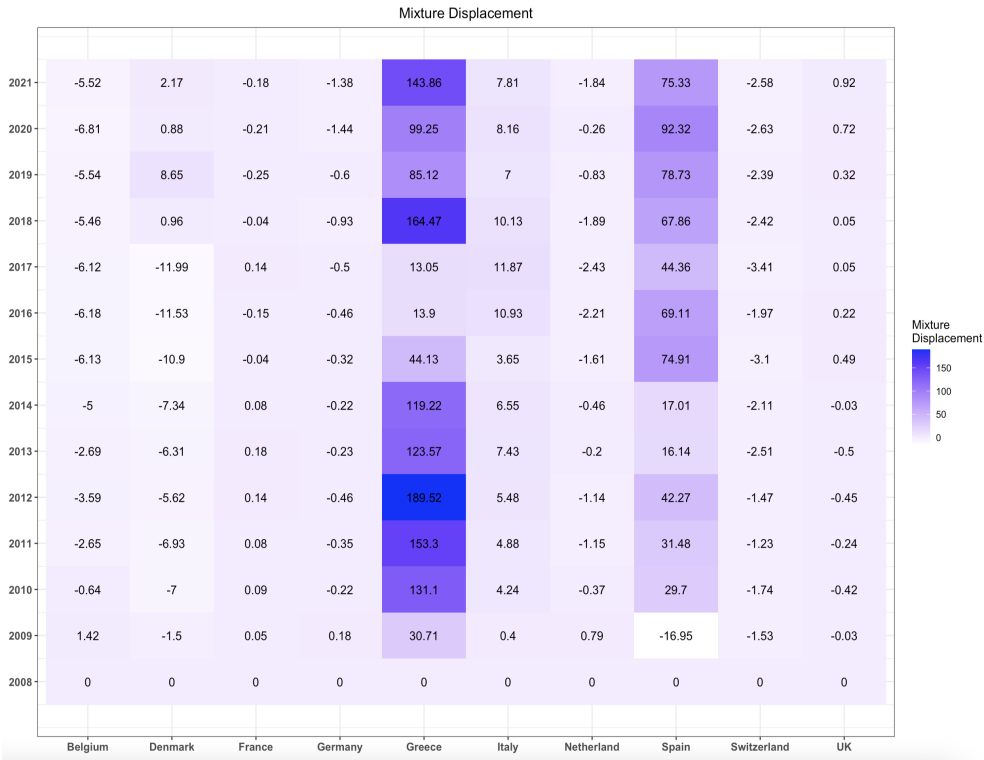
**Figure 2.** Kaniadakis Divergence on Compositional Data

#### 3.7.2. Mixture Displacement

The equation (19) calculates the mixture displacement. From the Figure (3), the mixture displacement for Greece and Spain is very high for all the years. The value for Spain in 2009



was less than zero—the only negative value for Spain. On the contrary, all other countries do not have too many high values.



**Figure 3.** Mixture Displacement on Compositional Data by taking 2008 as reference

3.7.3. Exponential Displacement

178

The equation (20) computes the exponential displacement. Figure (4) is the empirical result of equation (20). Like mixture displacement, while calculating exponential displacement, we can see that Spain and Greece have higher displacement than other European countries. The value for Spain in the year 2009 is meager.

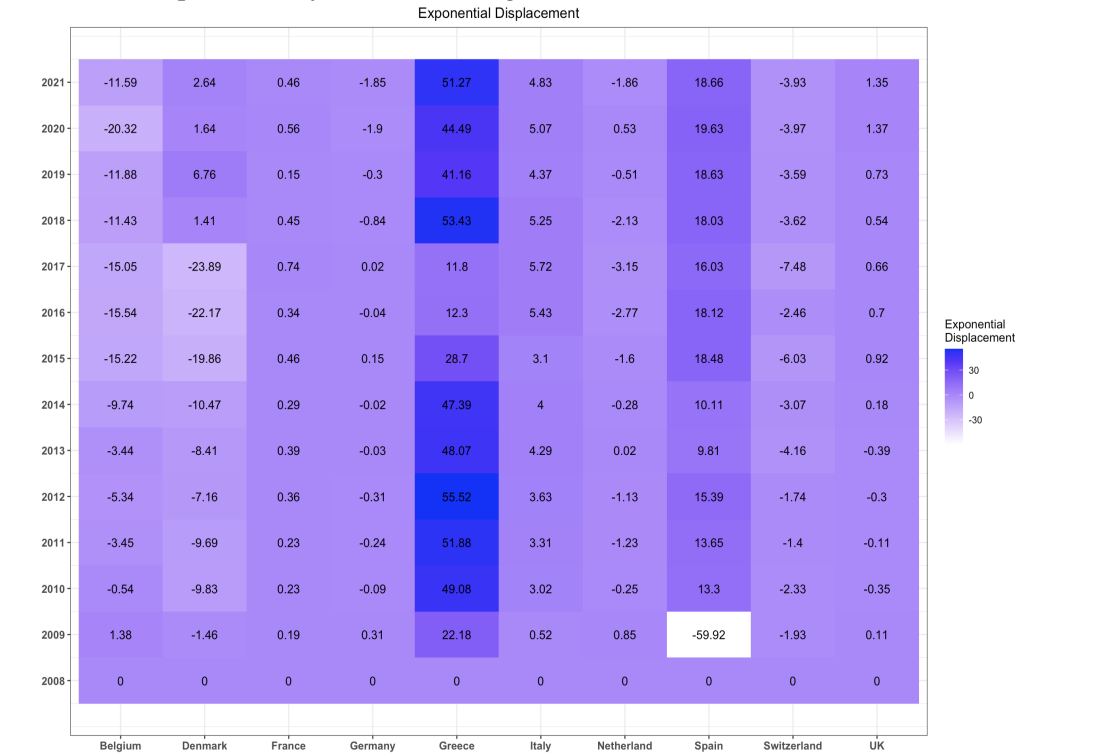


Figure 4. Exponential Displacement on Compositional Data by taking 2008 as reference

The computation of the affine displacement is a network computation; see the R suite or Python.

The computation of the exponential is a network equation.

FIND A GOOD EXAMPLE FOR ILLUSTRATION AND TEST

179

180

181

182

This appendix contains extra material, variants, and tentative redactions.

## Appendix A. Some Proofs

*Proof of eq. (5).*

Compute  $\log_K(\exp_K(x) \exp_K(y))$  using the algebraic expression of  $\log_K$  and  $\exp_K$ .

## Appendix B. Variations

*Appendix B.1. K-exponential form of a positive probability density*

If  $\Omega$  is a finite sample space, the probability simplex on  $\Omega$  is  $\mathcal{P}(\Omega)$ , and the open probability simplex is  $\mathcal{E}(\Omega)$ .

For all  $p \in \mathcal{E}(\Omega)$ , the function  $A \circ p$ ,  $A$  as in eq. (1), is strictly positive and proportional to a probability function  $\tilde{p}$ ,

$$\tilde{p} \propto A \circ p, \quad \text{namely,} \quad \tilde{p}(x) = \frac{A(p(x))}{\sum_x A(p(x))}.$$

The mapping

$$\mathcal{E}(\Omega) \ni p \mapsto \tilde{p} \in \mathcal{E}(\Omega)$$

is called the *escort* mapping, see [17]. We introduce a symbol for the escort expectation,

$$\tilde{\mathbb{E}}_p[u] = \mathbb{E}_{\tilde{p}}[u] = \frac{\sum_x u(x) A(p(x))}{\sum_x A(p(x))}.$$

For  $p, q \in \mathcal{E}(\Omega)$  write  $q$  as a deformed exponential model centred at  $p$ , we define the *Kaniadakis divergence* as  $\tilde{\mathbb{E}}_p[\log_K(p) - \log_K(q)] = \tilde{D}(p|q)$ . Clearly,  $\tilde{D}(p|p) = 0$ . If  $p \neq q$ , from eq. (6),

$$\tilde{D}(p|q) \propto \sum_x A(p(x))(\log_K(p) - \log_K(q)) > \sum_x (p(x) - q(x)) = 0.$$

Fix  $p \in \mathcal{E}(\Omega)$ . For all  $q \in \mathcal{E}(\Omega)$ , define

$$s_p(q) = (\log_K(q) - \log_K(p)) + \tilde{D}(p|q) = (\log_K(q) - \log_K(p)) - \tilde{\mathbb{E}}_p[(\log_K(q) - \log_K(p))].$$

If  $u = s_p(u)$ , then

$$q = \exp_K(u - \tilde{D}(p|q) + \log_K(p)), \quad \tilde{\mathbb{E}}_p[u] = 0. \quad (\text{A1})$$

Conversely, for all  $p \in \mathcal{E}(\Omega)$ , if  $u$  is a random variable such that  $\tilde{\mathbb{E}}_p[u] = 0$ , the real function

$$\mathbb{R}_+ \ni \psi \mapsto \sum_x \exp_K(u(x) - \psi + \log_K(p(x)))$$

is continuous, goes to 0 as  $\psi \rightarrow \infty$ , and, for  $\psi = 0$ , takes a value larger than 1 because of eq. (7):

$$\begin{aligned} \sum_x \exp_K(u(x) + \log_K(p(x))) &> \\ \sum_x \exp_K(\log_K(p(x))) + \sum_x A(\exp_K(\log_K(p(x))))u(x) &= \\ \sum_x p(x) + \sum_x A(p(x))u(x) &= 1. \end{aligned}$$

In conclusion, there exist a function

$$K_p: S_p = \left\{ u \in L(\Omega) \mid \tilde{\mathbb{E}}_p[u] = 0 \right\} \ni u \mapsto K_p(u) > 0,$$

such that

$$q = \exp_K(u - K_p(u) + \log_K(p)) \in \mathcal{E}(\Omega),$$

hence  $K_p(u) = \tilde{D}(p|q)$ . The mapping

$$s_p: \mathcal{E}(\Omega) \rightarrow B_p$$

is a bijection with inverse

$$e_p(u) = \exp_K(u - K_p(u) + \log_K(p)) \quad (\text{A2})$$

*Appendix B.2. Properties of the cumulant function  $K_p$ .*

Let us compute the derivatives of the function  $K_p$ . We use the notation

$$dK_p(u)[h] = \lim_{\theta \rightarrow 0} \theta^{-1} (K_p(u + \theta h) - K_p(u)) = \left. \frac{d}{d\theta} K_p(u + \theta h) \right|_{\theta=0}.$$

From eq. (10),

$$\begin{aligned} 0 &= \left. \frac{d}{d\theta} \sum_x \exp_K(u(x) + \theta h(x) - K_p(u + \theta h) + \log_K(p(x))) \right|_{\theta=0} = \\ &= \sum_x A(\exp_K(u(x) + \theta h(x) - K_p(u + \theta h) + \log_K(p(x))))(h(x) - \frac{d}{dt} K_p(u + \theta h)) \Big|_{\theta=0} = \\ &= \sum_x A(\exp_K(u(x) - K_p(u) + \log_K(p(x))))(h(x) - dK_p(u)[h]). \end{aligned}$$

It follows that for each  $p \in \mathcal{E}(\Omega)$ ,  $u, h \in B_p$ , it holds

$$dK_p(u)[h] = \tilde{\mathbb{E}}_q[h], \quad \text{where } q = e_p(u) \text{ see eq. (10)}. \quad (\text{A3})$$

Similarly, one should compute  $d^2 K_p$  and the convex conjugate of  $K_p$ . See [22], [23], and M. S. notes.

## References

1. Pawlowsky-Glahn, V.; Egozcue, J.J.; Tolosana-Delgado, R. *Modelling and Analysis of Compositional Data*; John Wiley & Sons, Ltd, 2015. <https://doi.org/10.1002/9781119003144>.
2. Pawlowsky-Glahn, V.; Egozcue, J.J. Compositional data and their analysis: an introduction. *Geological Society, London, Special Publications* **2006**, *264*, 1–10.
3. Egozcue, J.J.; Pawlowsky-Glahn, V. Compositional data: the sample space and its structure. *Test* **2019**, *28*, 599–638.
4. Coenders, G.; Ferrer-Rosell, B. Compositional data analysis in tourism: review and future directions. *Tourism Analysis* **2020**, *25*, 153–168.
5. Fiori, A.M.; Porro, F. A compositional analysis of systemic risk in European financial institutions. *Annals of Finance* **2023**, pp. 1–30.
6. Linares-Mustarós, S.; Coenders, G.; Vives-Mestres, M. Financial performance and distress profiles. From classification according to financial ratios to compositional classification. *Advances in Accounting* **2018**, *40*, 1–10.
7. Boonen, T.J.; Guillen, M.; Santolino, M. Forecasting compositional risk allocations. *Insurance: Mathematics and Economics* **2019**, *84*, 79–86.
8. Grifoll, M.; Ortego, M.; Egozcue, J. Compositional data techniques for the analysis of the container traffic share in a multi-port region. *European Transport Research Review* **2019**, *11*, 1–15.
9. Aitchison, J. *The statistical analysis of compositional data*; Monographs on Statistics and Applied Probability, Chapman & Hall, London, 1986; pp. xvi+416. <https://doi.org/10.1007/978-94-009-4109-0>.
10. Stolbov, M.; Shchepeleva, M. Systemic risk in Europe: Deciphering leading measures, common patterns and real effects. *Annals of Finance* **2018**, *14*, 49–91.
11. Dzhagityan, E.; Orekhov, M. Global Systemically Important Banks: Do They Still Pose Risks for Financial Stability?

12. Acharya, V.V.; Richardson, M.P.; et al. *Restoring financial stability: how to repair a failed system*; Vol. 542, John Wiley & Sons, 2009. 231
13. Acharya, V.; Engle, R.; Richardson, M. Capital shortfall: A new approach to ranking and regulating systemic risks. *American Economic Review* **2012**, *102*, 59–64. 232
14. Engle, R. Systemic risk 10 years later. *Annual Review of Financial Economics* **2018**, *10*, 125–152. 234
15. Engle, R.; Jondeau, E.; Rockinger, M. Systemic risk in Europe. *Review of Finance* **2015**, *19*, 145–190. 235
16. Amari, S.i. *Information geometry and its applications*; Vol. 194, *Applied Mathematical Sciences*, Springer, [Tokyo], 2016; pp. xiii+374. 236
17. Naudts, J. Generalised exponential families and associated entropy functions. *Entropy* **2008**, *10*, 131–149. <https://doi.org/10.3390/entropy-e10030131>. 237
18. Kaniadakis, G. Non-linear kinetics underlying generalized statistics. *Physica A* **2001**, *296*, 405–425. 238
19. Kaniadakis, G. H-theorem and generalized entropies within the framework of nonlinear kinetics. *Physics Letters A* **2001**, *288*, 283–. 239
20. Pistone, G.; Riccomagno, E.; Wynn, H.P. *Algebraic statistics: Computational commutative algebra in statistics*; Vol. 89, *Monographs on Statistics and Applied Probability*, Chapman & Hall/CRC, Boca Raton, FL, 2001; pp. xvii+160. 240
21. Montrucchio, L.; Pistone, G. A Class of Non-parametric Deformed Exponential Statistical Models. In *Geometric Structures of Information*; Springer International Publishing, 2018; pp. 15–35. [https://doi.org/10.1007/978-3-030-02520-5\\_2](https://doi.org/10.1007/978-3-030-02520-5_2). 241
22. Pistone, G. Information Geometry of the Probability Simplex: A Short Course. *Nonlinear Phenomena in Complex Systems* **2020**, *23*, 221–242. 242
23. Chirco, G.; Pistone, G. Dually affine Information Geometry modeled on a Banach space, 2022. arXiv:2204.00917, <https://doi.org/10.48550/ARXIV.2204.00917>. 243
24. Kaniadakis, G. Statistical mechanics in the context of special relativity. *Physical Review E* **2002**, *66*, 056125 1–17. 244
25. Kaniadakis, G. Statistical mechanics in the context of special relativity. II. *Phys. Rev. E* **2005**, *72*, 036108. <https://doi.org/10.1103/PhysRevE.72.036108>. 245
26. Pistone, G.  $\kappa$ -exponential models from the geometrical viewpoint. *The European Physical Journal B Condensed Matter Physics* **2009**, *71*, 29–37. <https://doi.org/10.1140/epjb/e2009-00154-y>. 246
27. Montrucchio, L.; Pistone, G. Deformed Exponential Bundle: The Linear Growth Case. In *Lecture Notes in Computer Science*; Springer International Publishing, 2017; pp. 239–246. [https://doi.org/10.1007/978-3-319-68445-1\\_28](https://doi.org/10.1007/978-3-319-68445-1_28). 247
28. Amari, S.; Nagaoka, H. *Methods of information geometry*; American Mathematical Society, 2000; pp. x+206. Translated from the 1993 Japanese original by Daishi Harada. 248
29. Weyl, H. *Space- time- matter / by Hermann Weyl*; Dover: New York, 1952. translation of the 1921 RAUM ZEIT MATERIE. 249